



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CAPSTONE PROJECT

AI – BASED LOAN ELIGIBILITY PREDICTION AND RISK ASSESSMENT SYSTEM

CSA0905 – JAVA PROGRAMMING

UNDER THE SUPERVISION OF:

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PROBLEM BACKGROUND

- **Application Domain:**
Banking and Financial Services Loan Processing and Credit Risk Management
- **Current Scenario:**
Many financial institutions still rely on manual verification and traditional credit assessment methods. Loan approval often requires significant paperwork and human intervention, leading to delays. Human judgment may result in inconsistent loan decisions.
- **Need for the Project :**
Automate loan eligibility evaluation, Reduce loan processing time, Improve the accuracy of credit risk assessment, Minimize financial losses due to bad loans.
- **Target Users:**
Banks Financial Institutions, Credit Unions Loan Officers, Customers applying for loans.

PROBLEM STATEMENT

- **Problem Definition**

Financial institutions require an intelligent system to evaluate customer loan applications efficiently while minimizing credit risk. Traditional methods are time-consuming and may produce inconsistent decisions.

- **Evidence / Statistics**

According to industry reports, non-performing loans (NPLs) remain a major challenge for banks worldwide. Manual loan verification increases processing time and operational costs. Credit risk is one of the leading causes of financial losses in lending institutions.

- **Limitations of Current Systems**

Manual verification process, Longer approval time, Human errors in decision-making, Inconsistent loan approvals, Limited real-time risk assessment.

- **Expected Impact**

Faster loan processing, Improved decision accuracy, Reduced credit risk, Better customer satisfaction, Increased operational efficiency

OBJECTIVE

- **Primary Objective**

To develop a Java-based AI Loan Eligibility Prediction and Risk Assessment System that automatically evaluates customer loan applications based on financial and credit-related information.

- **Measurable Objectives**

Develop a customer profile management module. Predict customer credit risk using predefined intelligent rules. Automatically recommend loan approval, rejection, or further review. Store and manage customer and prediction records using SQLite. Provide a dashboard to display customer details, risk level, and loan decisions.

- **Project Scope**

Customer registration and profile management, Credit risk prediction ,Automated loan eligibility decision, Loan decision dashboard, Data storage using SQLite database, Java Swing-based desktop application

LITERATURE SURVEY

S.No.	Paper / Year	Method / Technique Used	Key Contribution	Limitation
1	Loan Prediction using Decision Tree (2023)	Decision Tree	Fast and easy to implement. Provides interpretable results.	Lower accuracy on complex and imbalanced data.
2	Credit Risk Assessment using Random Forest (2023)	Random Forest	High prediction accuracy and handles large feature sets well.	Computationally expensive and less interpretable.
3	Machine Learning for Loan Approval (2024)	Support Vector Machine (SVM)	Effective in high-dimensional spaces and works well for small datasets.	Sensitive to parameter tuning and kernel selection.
4	Credit Scoring using Neural Networks (2024)	Artificial Neural Network (ANN)	Captures complex non-linear relationships in data.	Requires large datasets and higher training time.
5	Explainable AI for Loan Decisions (2025)	Explainable AI (XAI) Techniques	Improves transparency and trust in loan decision systems.	Limited real-world adoption and standardization.

Existing System vs Proposed System

Feature	Existing System	Proposed System	Advantage
Loan Verification	Manual loan verification	Automated loan eligibility prediction	Faster loan processing with reduced manual effort
Credit Assessment	Basic credit score check	Multi-factor credit risk assessment	Improves prediction accuracy and reliability
Decision Making	Slow and time-consuming approval process	AI-inspired rule-based decision making	Faster and consistent loan decisions
Accuracy	Human errors possible during verification	Intelligent and consistent risk evaluation	Reduces errors and improves decision quality
Data Management	Separate record management	Centralized SQLite database	Secure, organized, and quick data access

REQUIREMENTS

- **Functional Requirements:**

Customer registration and profile management, Credit risk prediction, Automatic loan eligibility decision, Loan decision dashboard.

- **Non-Functional Requirements:**

Fast response time, User-friendly interface, Secure and reliable data storage, Easy to maintain and scalable.

- **Hardware Requirements:**

Intel Core i3 or above 4 GB RAM (minimum) 500 MB free storage Windows/Linux OS

- **Software Requirements:**

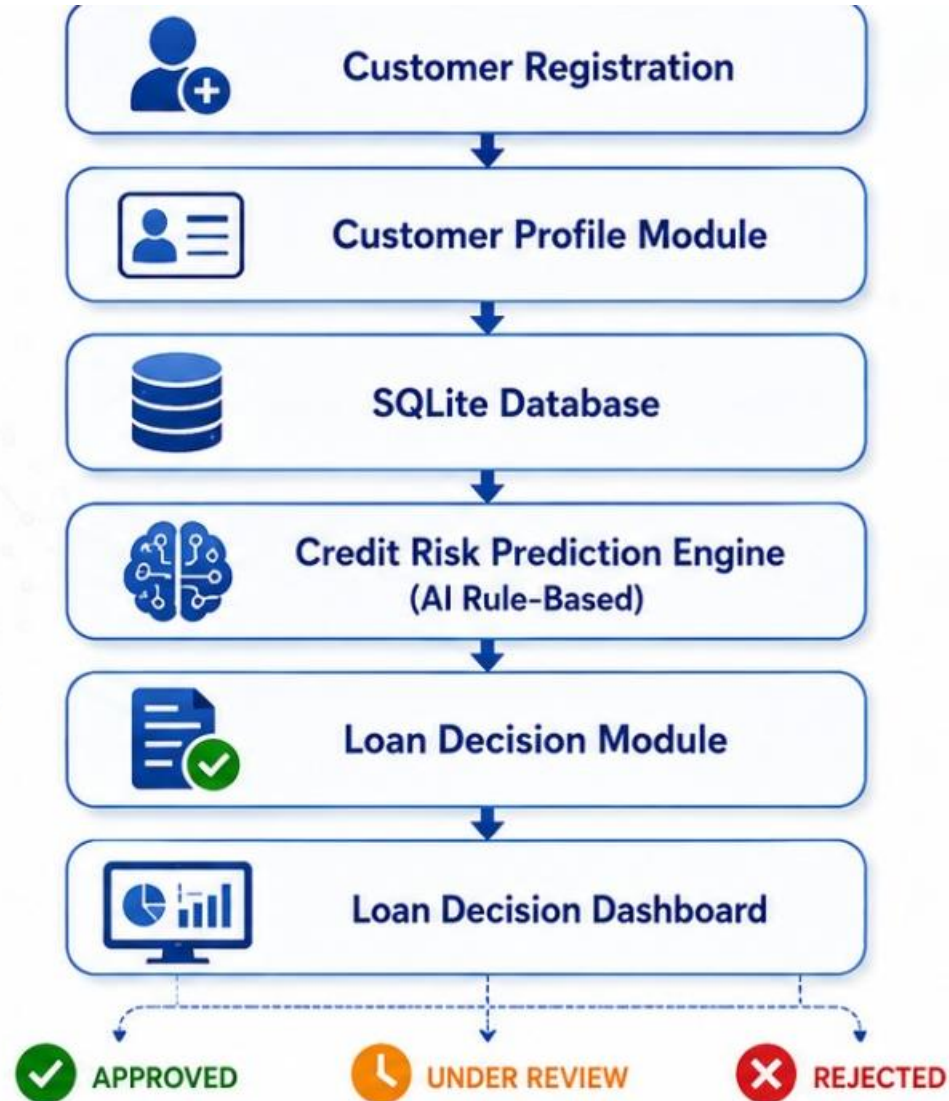
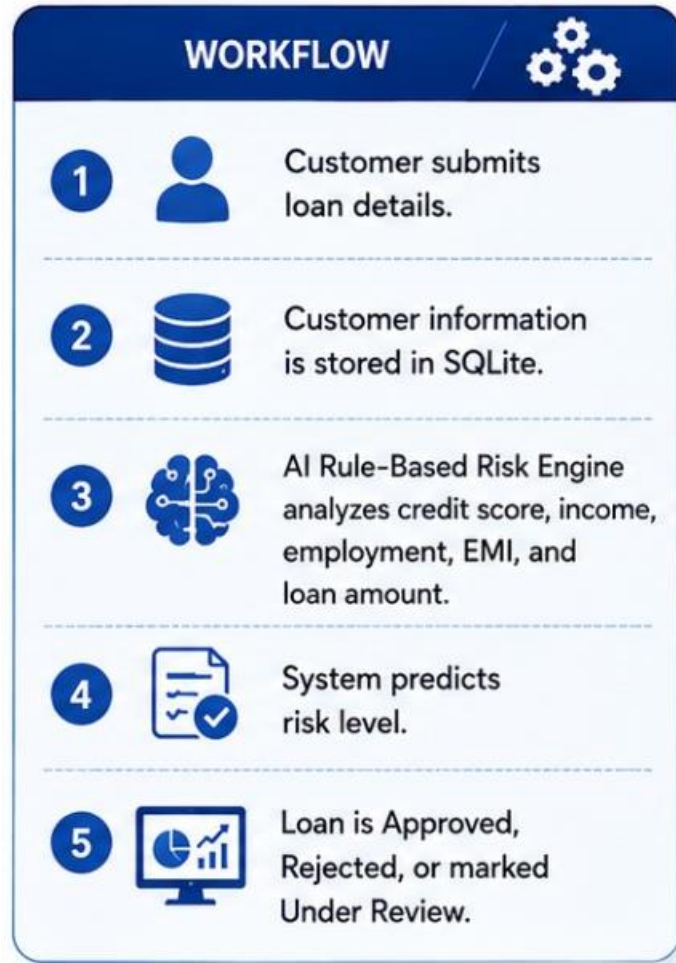
Java (JDK 25 or above) IntelliJ IDEA / Eclipse SQLite Database JDBC Driver Maven (Optional) Dataset / APIs Dataset: Sample customer loan data (Income, Credit Score, Employment, EMI, Loan Amount, Existing Loan) APIs: Not required (Standalone Java Swing application)



Engineer to Excel



SYSTEM ARCHITECTURE





PLANNING



1 Work Breakdown Structure (WBS)



AI-Based Loan Eligibility Prediction and Risk Assessment System

- Requirement Analysis
- Database Design (SQLite)
- GUI Development (Java Swing)
- Customer Profile Module
- Credit Risk Prediction Module
- Loan Decision Dashboard
- Testing & Debugging
- Documentation & Presentation

2 Gantt Chart (6 Weeks)

Task	W1	W2	W3	W4	W5	W6
Requirement Analysis	■					
Database Design (SQLite)	■	■				
GUI Development (Java Swing)		■	■			
Module Development			■	■		
Testing				■	■	
Documentation & PPT					■	■

4 Risk Analysis

	Risk	Mitigation
⚠	Incorrect Data	Input Validation
🗄	Database Failure	Regular Backup
🧠	Prediction Errors	Improve Decision Rules
📅	Schedule Delay	Weekly Progress Monitoring

3 Milestones



Requirement Analysis



Database & UI Completed



Risk Prediction Module



Dashboard Developed



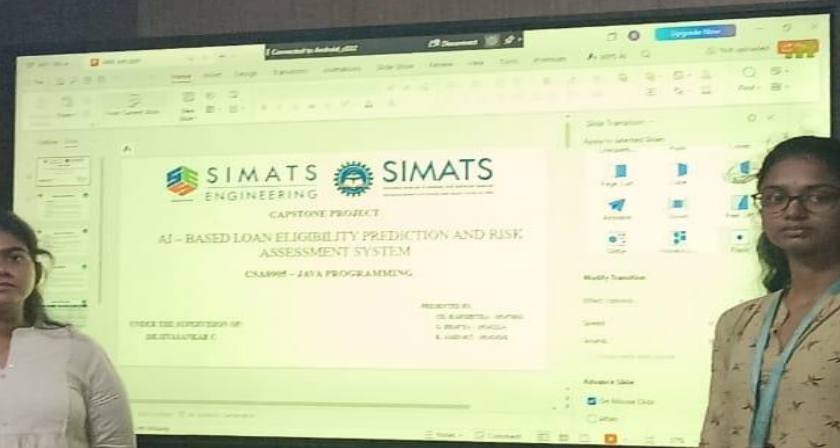
Testing & Final Presentation



EXPECTED OUTPUT

- **Project Deliverables:** Java Swing desktop application, SQLite database ,Credit risk prediction module, Loan decision dashboard, Project report and presentation
- **Benefits:** Faster loan processing, Reduced manual effort, Improved credit risk assessment, Consistent loan decisions, User-friendly and low-cost solution.
- **References:**
 - A. K. Sharma and R. Gupta, "Machine Learning Approaches for Loan Approval Prediction," IEEE Access, vol. 12, pp. 14567–14578, 2024.
 - S. Patel and M. Verma, "Credit Risk Assessment Using Artificial Intelligence," Proc. IEEE Int. Conf. on AI and Data Engineering, pp. 120–126, 2023. J.
 - Brown and L. Wang, "Explainable AI for Financial Decision Making," IEEE Transactions on Artificial Intelligence, vol. 5, no. 2, pp. 210–220, 2025.
 - R. Singh et al., "Loan Eligibility Prediction Using Decision Tree Algorithms," IEEE Int. Conf. on Computing and Analytics, pp. 98–104, 2024.

THANK YOU



 GPS Map Camera

Mevalurkuppam, Tamil Nadu, India 
Eee Department, Mevalurkuppam, Tamil Nadu 602105, India
Lat 13.026742° Long 80.016104°
Friday, 07/08/2026 09:28:25 AM GMT +05:30